CAPSTONE 31 — Documented AI Prompt Library (Skills 1–8)

10 Structured R-C-T-V Prompts Across ChatGPT, Gemini, Claude, DeepSeek & NotebookLM

1. Prompt Engineering Architecture for DevOps & Git Education

Every prompt in this library uses the **R-C-T-V Framework** (*Role, Context/Audience, Task Constraints, Verification Criteria*) to minimize AI hallucinations and tailor outputs specifically to 2nd- and 3rd-year CSE student project teams across all 8 required AI skills.

Prompt ID & Skill	AI Tool Used	Exact Documented Prompt	Verification & Output Outcome
P-01 Skill 1: Prompting & Audience Calibration	ChatGPT & Gemini	*"Act as a university CSE DevOps lab instructor. Diagnose the top 5 Git and integration failure modes experienced by 4-student 2nd- and 3rd-year undergraduate project teams who only know basic git add/commit/push. Contrast their habits with industry GitHub Flow."*	Identified concrete student pain points (overwriting `main`, panic on merge conflicts, committing `.env` files) used to structure the Research Report.
P-02 Skill 2: Technical Explanation	ChatGPT & Claude	*"Explain Git branching, HEAD pointers, Fast-Forward merges, and Three-Way merges to a 2nd-year CSE student who just completed Data Structures (linked lists and directed acyclic graphs). Use a DAG node-pointer analogy and avoid vague metaphors. Cite Pro Git Ch. 3."*	Produced an accurate DAG pointer explanation connecting commits to immutable nodes with parent pointers, verified against Pro Git Ch. 3.1.
P-03 Skill 2: Workflow Comparison	Google NotebookLM & Gemini	*"Using only the official GitHub Flow documentation and Pro Git Chapter 3.4, compare GitHub Flow, Gitflow, and Trunk-Based Development in a structured Markdown table evaluating branch overhead, CI requirements, and suitability for a 14-week university capstone team."*	Grounded comparison proving GitHub Flow provides the optimal balance of safety and low overhead for student teams.

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P-04 Skill 4: Documentation (Command Guide)	Claude & Gemini	*"Create a task-oriented Git command cheat sheet for student teams using modern Git 2.23+ commands (prefer git switch and git restore over overloaded git checkout). Cover setup, branching, staging, syncing with main, conflict resolution, and safe undo (git revert vs git reset)."*	Eliminated legacy `git checkout` ambiguity and added explicit warnings against `git push --force` on shared branches.
P-05 Skill 4: Documentation (PR Template)	Claude	*"Draft a .github/pull_request_template.md file for a CSE student team repository. Include sections for Summary, Linked Issue (Closes #), Type of Change (Conventional Commits), Local Test/Lint Verification checkboxes, and a Responsible AI & Secret-Hygiene checklist."*	Deployed directly to `.github/pull_request_template.md` in the sample repository; verified against GitHub PR Template Docs.
P-06 Skill 3 & 7: CI Workflow & Code Verification	Gemini & DeepSeek	*"Write a production-grade .github/workflows/ci.yml file for a Python repository that runs on push and pull_request to main. Use a matrix for Python 3.11 and 3.12, cache pip dependencies, run flake8 static analysis and pytest unit tests, and enforce least-privilege GITHUB_TOKEN permissions. Use current v4/v5 actions."*	Cross-checked against GitHub Python CI Docs; confirmed `actions/checkout@v4`, `actions/setup-python@v5`, and `permissions: contents: read`.
P-07 Skill 7: Automation (Git Hook)	Claude & DeepSeek	*"Write a zero-dependency Python 3 Git commit-msg hook script (scripts/commit-msg) that: (1) validates the commit subject line against Conventional Commits v1.0.0 regex, and (2) scans git diff --cached for accidental API keys (AWS, GitHub PAT, Google AIza, OpenAI sk-) and hardcoded passwords, exiting non-zero with clear student remediation advice on failure."*	Tested in sandbox against invalid headers (`fixed bug` → exit 1) and valid headers (`feat(analytics): ...` → exit 0).

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P-08 Skill 5: Quiz Design	Gemini & ChatGPT	*"Design a 10-question multiple-choice quiz for 2nd/3rd-year CSE students covering Git branching pointers, 3-way merges, PR templates, branch protection status checks, flake8 vs pytest in GitHub Actions, commit-msg hooks, AI YAML hallucinations, and secret rotation ethics. Include plausible distractors based on real student misconceptions and a detailed answer key."*	Compiled into `07_Quiz_and_Answer_Key.pdf` with 100% source-verified rationales.
P-09 Skill 6: Presentation Design	Gemini & Gamma (Structure)	*"Outline a 6-slide executive and instructional presentation deck for CSE project teams titled 'DevOps and Git Workflow Primer with CI Basics'. Use declarative assertion-evidence slide titles, limit each slide to 3 concise bullet fragments, and include a quantitative Before/After impact chart."*	Built and validated via `python-pptx` into `06_Presentation_Deck.pptx` with 0 layout or legibility errors.
P-10 Skill 3 & 8: Adversarial Hallucination & Security Audit	Gemini, Claude & NotebookLM	*"Act as a DevSecOps auditor. Review this naive AI-generated GitHub Actions YAML [pasted naive workflow using pull_request_target, actions/checkout@v2, ::set-output, and unquoted ${{ github.event.pull_request.title }} in a run block]. Identify every deprecated syntax, script injection vulnerability, and permission flaw, citing GitHub Docs Secure Use Reference."*	Documented 5 critical AI failure modes and their exact remediations in the Verification & Ethics Report.